

BILAL AHAMAD

Lead Embedded Firmware & Systems QA Engineer

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SUMMARY

Quality engineering leader with 18+ years validating firmware and embedded systems for Amazon (Alexa devices), Google (VR), Rivian, Cruise, and Samsara. Builds the test architectures behind safety-critical and AI/ML-driven products — hardware-in-the-loop rigs, CI/CD device labs, and automation frameworks that have delivered \$3M+ in cost savings and \$2.1M in new revenue. Currently architecting 0-to-1 validation for a sensor-integration IDE and AI-augmented test pipelines.

CORE SKILLS

Test & Automation: Python, test framework architecture, hardware-in-the-loop (HIL), CI/CD (Jenkins), Docker/QEMU, Appium, Selenium, ADB & Android, QNX/C++

Domains: Firmware & IoT validation, automotive infotainment & V2X, AI/ML system testing, video quality, Bluetooth/Wi-Fi (WFA) certification, OTA updates

AI-Augmented Engineering: Claude Code, Cursor, Gemini — AI pair-programmed test frameworks, agentic automation pipelines

Software & Data: TypeScript, Node.js, React/Next.js, Pandas, Plotly & Splunk dashboards, AWS, GitHub Actions

EXPERIENCE

Founding System Architect & Technical QA Lead — [Stealth Mode](#) Sep 2025 – Present · Remote

0-to-1 architecture and validation of a Unified Sensor Integration IDE automating OEM dependency integration.

- Engineered a pre-validated compatibility layer for industry-standard chips (ARM, ESP32) and sensors (LiDAR/Radar), reducing manual bring-up cycles by ~70%.
- Built a containerized Docker/QEMU simulation ecosystem enabling high-fidelity A/B validation and performance engineering before physical hardware deployment.

Senior Firmware Quality Lead — [Samsara Inc](#) Dec 2023 – Jul 2025 · San Francisco, CA

E2E system test strategy for the AI dash-cam product series in an AI/ML video-based safety ecosystem.

- Reduced regression cycles by 50% and improved release cadence through network-reliability and connectivity automation.
- Built automated pipelines validating video quality, AI/ML algorithm accuracy, and workload performance.

Senior Automation Engineer II — [Cruise LLC](#) Oct 2022 – Jun 2023 · San Francisco, CA

Owned test delivery for autonomous-vehicle computational systems and safety-critical firmware releases.

- Developed the automation framework for sanity, integration, and stress testing, achieving 75% test coverage.
- Led fault-tolerance, redundancy, and network reliability test design (V2X, PCAP replay).

Infotainment & V2X Firmware Test Lead — [Rivian Automotive](#) Jun 2021 – Sep 2022 · Palo Alto, CA

Directed testing strategy and E2E delivery for in-car infotainment across R1T, R1S, and Fleet models.

- Designed the QNX + Android automation framework covering OTA updates, UI/UX, and navigation through continuous releases.
- Achieved critical KPI goals from out-of-box experience through the full user-flow lifecycle.

Senior QA Engineer II & Test Lead — [Amazon Lab126](#)

Jun 2018 – Jun 2021 · Sunnyvale, CA

QA owner for Alexa Voice Service products: Echo Buds (1st & 2nd Gen), Echo Auto, and Dash Button.

- Automated the Snippet API library, cutting manual testing costs by \$3M.
- Stood up CI/CD test labs running 50+ devices continuously — 30% fewer post-launch issues.
- Built Splunk live dashboards for trend and metric visibility, improving test efficiency to 80%.

Senior Test Engineer — [Tech Mahindra / Google](#)

Jan 2016 – Jun 2018 · Mountain View, CA

Led system testing for Google's VR controllers, shipping 1st & 2nd Gen controllers and Daydream View.

- Engineered a 3DOF Arduino robotic arm for automated motion-tracking testing, reducing execution hours by 80%.
- Developed Python tooling for UART battery measurement and 2DOF motion-tracking evaluation.

Product Validation Engineer — [Cognizant / Cisco](#)

Sep 2015 – Jan 2016 · Boston, MA

- Built a Python test framework for set-top-box streaming apps simulating remote-key events — test scope +40%, execution time halved.

Radio Validation Engineer — [Wistron Mobile Solutions](#)

Dec 2014 – Sep 2015 · Rolling Meadows, IL

- Re-architected the WLAN interoperability automation framework with Android support — \$1.3M in savings; led Wi-Fi Alliance pre-certification testing.

Senior Software Engineer & Test Automation Lead — [L&T Infotech / Motorola Mobility](#)

Oct 2009 – Dec 2014 · Chicago, IL

- Pioneered Bluetooth qualification test automation for Motorola handsets, generating \$2.1M in revenue.
- Devised a 3-tier automation infrastructure — 3x code reuse, 70% less test-preparation effort.

Software Developer — [Luminous Infoways](#)

Oct 2008 – Sep 2009 · Bhubaneswar, India

- Functional and integration testing for web portal applications; deployment and feature integration.

SELECTED PROJECTS

US Live Layoff Monitoring Dashboard — automated pipeline unifying WARN notices from 46 states + DC into one dataset, JSON API, and alert system; 100% automated, twice-daily. [bilalahamad0.github.io/warn](https://github.com/bilalahamad0/warn)

Adhan Caster — Cross-Browser Extension — Manifest V3 extension live on the Chrome, Firefox, and Edge stores; auto-pauses media across all tabs; 194 tests / 14 suites. github.com/bilalahamad0/adhan-ce

Smart-Home IoT Media Caster — Raspberry Pi + Android TV (ADB) orchestration layer, 10 microservices; 3-week cycle compressed to 4 days with AI pair programming. github.com/bilalahamad0/adhan-api

CERTIFICATIONS

- Google AI Professional (2026)
- Google Project Management Professional (2026)
- ISTQB Certified Tester Foundation Level (CTFL)
- Software Testing Foundations: Integrating AI into Quality Process (2026)
- AI Coding Agents with GitHub Copilot and Cursor (2025)
- Scrum: Advanced (2021)

EDUCATION

B.Tech, Electronics & Telecommunications Engineering

Biju Patnaik University of Technology — 2004 – 2008